Computer Science & IT



C Programming

Data Types and Operators

Lecture No. 06



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Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Arithmetic

Relational operator

Logical Operator

22

11

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Topics to be Covered

gate questions



- Topic: Increment & Decrement Operator
- Topic: Bitwise Operator
- Topic
- Topic
- Topic

Slide



Toipc:Question

```
#include <stdio.h>
   int main () {
  int a =(( 5+6!=10 )==8)&&((6*2/4>10)==(10>6));
   printf("%d", a);
  }
```

The output of the program is 0

((11!=10)==8)

1==8

(6*2/4>10)==(10>6))

22 operahon

1 1 - 1

0

Slide



Toipc:Question

```
#include <stdio.h>
    int main ()   {
    int a =(( 5+6!=10 )==8)&&((6*2/4>10)==(10>6));
    printf("%d", a);
    }
```

The output of the program is _____

Answer :0

Slide



Toipc:Question

```
#include <stdio.h>
    int main ()  {
    int a =(( 5+6!=10 )==8)||((6*2/4>10)==(10>6));
    printf("%d", a);
    }
```

The output of the program is _____



Toipc:Question

```
#include <stdio.h>
    int main ()  {
    int a =(( 5+6!=10 )==8)||((6*2/4>10)==(10>6));
    printf("%d", a);
    }
```

The output of the program is __0___

Slide



Toipc:Question

```
#include <stdio.h>
      int main ()   {
          int a =5+5!=10|| 6+4;
       printf("%d", a);
     }
```

The output of the program 1

Non- zero

5+5 != 10 || 6+4

10 != 10 || 10

0 || 10 ← Non zero

0 || 1 = 1

Slide

Zero

−2

↑

Non zero

Non Zero - true ✓

Zero - value - false

−2 || 0 = 1 || 0 = 1



Toipc:Question

```
#include <stdio.h>
     int main ()  {
        int a =5+5!=10|| 6+4;
     printf("%d", a);
    }
```

The output of the program _1__

Slide



Toipc: Short Circuit Code

Short circuit if logical operator && is applied if first operand is 0 then second operand is Not evaluated

0 && [ ] will Not execute

Slide



Toipc: Short Circuit Code

In logical operator || (OR) if first operand is 1 then Second operand will not execute.

1 || [ ]

this expression will Not execute.

Slide



Toipc: Increment & decrement Operator





Slide



Toipc: Increment & decrement Operator

Modify the content of a variable

$X = X + 1$

int x = 10

++x
pre increment

x++
post increment

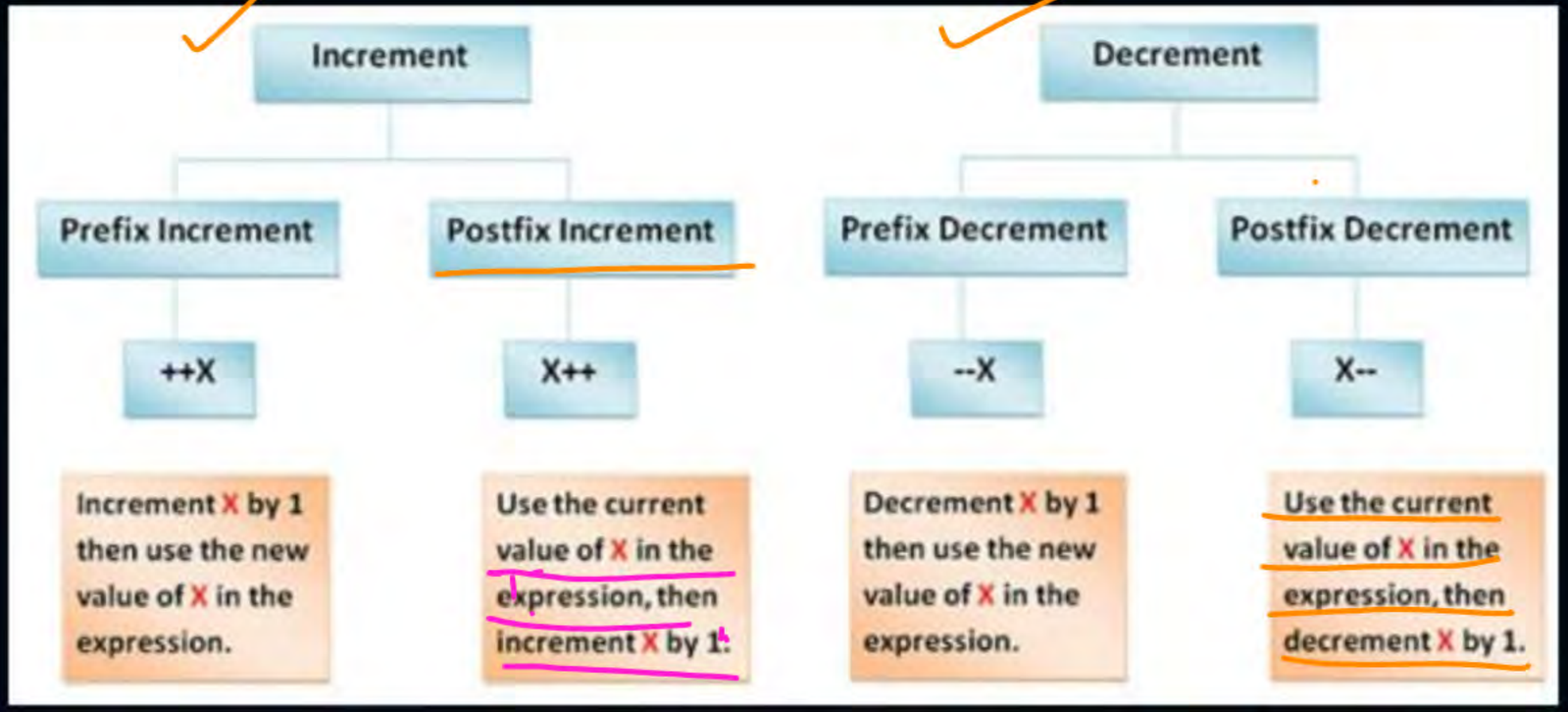
--x
pre decrement

x--
post decrement

Slide



Toipc: Increment & decrement Operator



Slide



Toipc: Question

```
#include <stdio.h>
      int main ()  {
       int x=5;
       int y;
       y = ++x;
      printf("%d %d", x, y);
     }
```

6 6

x =

preIncrement

y = ++x; x = ~~5~~ 6

y = 6

The output of the program

(A) 5,5

(B) 5,6

(C) 6,5

(D) 6,6

Slide

First Increment

$y = ++x$

$y = x++$

assign then Increment



Toipc: Question

```
#include <stdio.h>
     int main ()  {
      int x=5;
      int y;
      y = ++x;
     printf("%d %d", x, y);
    }
```

The output of the program

(A) 5,5 (B) 5,6 (

C) 6,5 (D) 6,6

Answer

D

Slide



Toipc: Question

What is the output of the program?

```
#include <stdio.h>
      int main ()  {
       int x=5;
       int y;
       y = x++;
      printf("%d %d", x, y);
     }
```

Output of the program is_____

(A) 5,5

(B) 5,6

(C) 6,5

(D) 6,6

x=5 → post

y = x++

y = 5 = y 5

x = 6

Slide



Toipc: Question

What is the output of the program?

```
#include <stdio.h>
      int main ()  {
       int x=5;
       int y;
       y = x++;
      printf("%d %d", x, y);
     }
```

[C]

In post Increment
first assign then
Increment will happen

y = 5

x = 6

Output of the program is_____

(A) 5,5

(B) 5,6

(C) 6,5

(D) 6,6

Slide



Bit-wise Operator



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```
int a
int x=5
int y=6
a = ++x || ++y
printf("%d %d", x, y)
```

a = 6 || 6 = 6 || 6

Left to right

Logical OR

Non zero value (T)

6 || 6

6, 6

what program will print?

Slide

Q22

```
int x = 0;
int y = 6;
int a;
a = x++ 22 ++y;
printf("%d %d", x, y);
```

1,6

expression

x will be 0 after this expression

Short Circuit

1, 6



Number System

1. Decimal No. (0,1,2 .. 9) int a = 10;
   ↑ Decimal

2. Binary System (0, 1). Singed/unsigned

3. octal system int a = 012;
   ↑ is octal No.
   the No. begin with 0
   (0, 1, 2, 3, 4, 5, 6 7)

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int a = 013; ← octal No

printf("%d", a); = // 11

Digital

1

Binary

octal

Hexa

Detail

$a = 013$

$1 \times 8^1 + 3 \times 8^0 = 8 + 3 = 11$

```
int a = 0x20;
printf("%d", a);
```

0x20

1 0

2 0

$2 \times 16^1 + 0 \times 16^0$

$= 2 \times 16 + 0 = 32$

4. Hexadecimal (0, 1, 2 .. 9, A, B, C, D, E F)

16 bits

int a = 0x11

↑ Hexadecimal

Bitwise operator (6 bits)

AND 1. 2

OR 2. 1

Negation 3 ~

XOR 4 ^

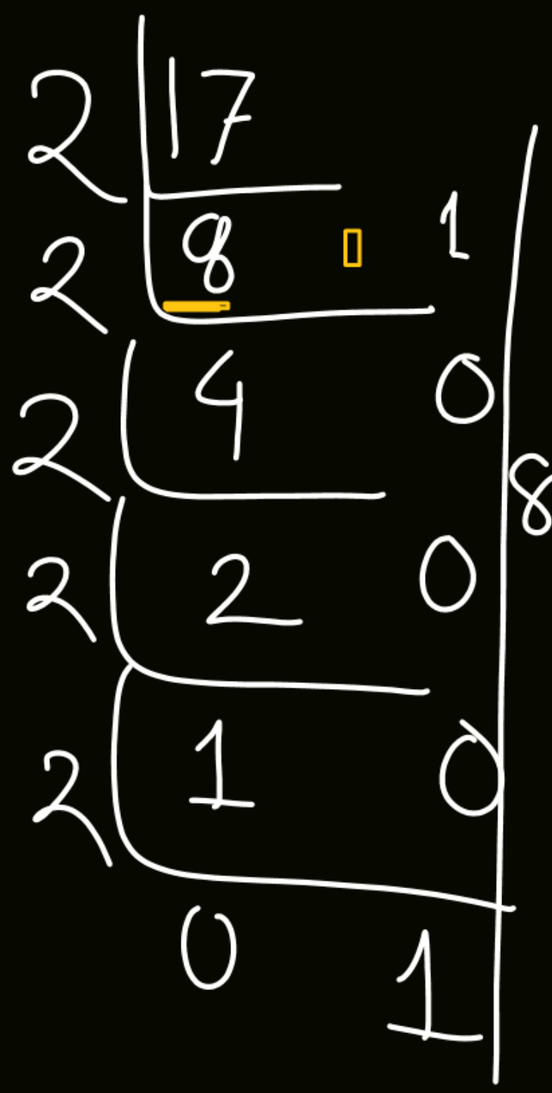
5. >> Right shift

6. << left shift



Bit-wise Operator

Operators	Meaning of operators
&	Bitwise AND
|	Bitwise OR
^	Bitwise exclusive OR
~	Bitwise complement
<<	Shift left
>>	Shift right



int x = 5;

int y = 17;

int z = x & y;

1 Byte 8 bits

AND

z = 1

8bit 5 - 00000101

17 - 00010001

00000001



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
      int x = 5, y=17, z
      z = x&y;
      printf("%d", z);
 }
```

(A) 1
(B) 21
(C) 2
(D) -6

z = 1

```
  0 0 0 0 0 1 0 1
  0 0 0 1 0 0 0 1
-----------------
  0 0 0 1 0 0 0 1
```

Slide



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
      int x = 5, y=17, z
      z = x&y;
      printf("%d", z);
 }
```

(A) 1 (B) 21

(C) 2 (D) -6

Slide



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
      int x = 5, y=17, z
      z = x|y;
      printf("%d", z);
 }
```

(A) 1
(B) 21
(C) 2
(D) -6

OR

00000101
00010001

00010101

7 6 5 4 3 2 1 0

$1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$= 16 + 4 + 1 = 21$

Slide



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
      int x = 5, y=17, z
      z = x|y;
      printf("%d", z);
 }
```

(A) 1 (B) 21

(C) 2 (D) -6

Slide



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
     int x = 5, y=17, z
      z = x^y;
       printf("%d", z);
 }
```

(A) 1

(B) 21

(C) 20

(D) -6

XOR

x	y	x^y
0	0	0
0	1	1
1	0	1
1	1	0

Slide

5 - 0 0 0 0 0 1 0 1

17 - 0 0 0 1 0 0 0 1

---

0 0 0 1 0 1 0 0

4 3 2 1 0

x	y	x^y
0	0	0
0	1	1
1	0	1
1	1	0

$1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$

$16 + 4 = \boxed{20}$



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
      int x = 5, y=17, z
      z = x^y;
      printf("%d", z);
 }
```

(A) 1
(B) 21
(C) 20
(D) -6



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
        int x = 5, z
        z = ~x;
        printf("%d", z);
    }
```

(A) 1
(B) 21
(C) 20
(D) -6

Slide



Bit-wise Operator

What is the output the program

```
#include <stdio.h>
    int main ()  {
        int x = 5, z
        z = ~x;
        printf("%d", z);
    }
```

(A) 1

(B) 21

(C) 20

(D) -6

Slide



Bit-wise Operator

Left shift

<<

```
#include<stdio.h>
   int main(){
            char a = 8;
            int k;
            k =a<<3;
            printf("%d", k);
            return 0;
   }
```

a <<3

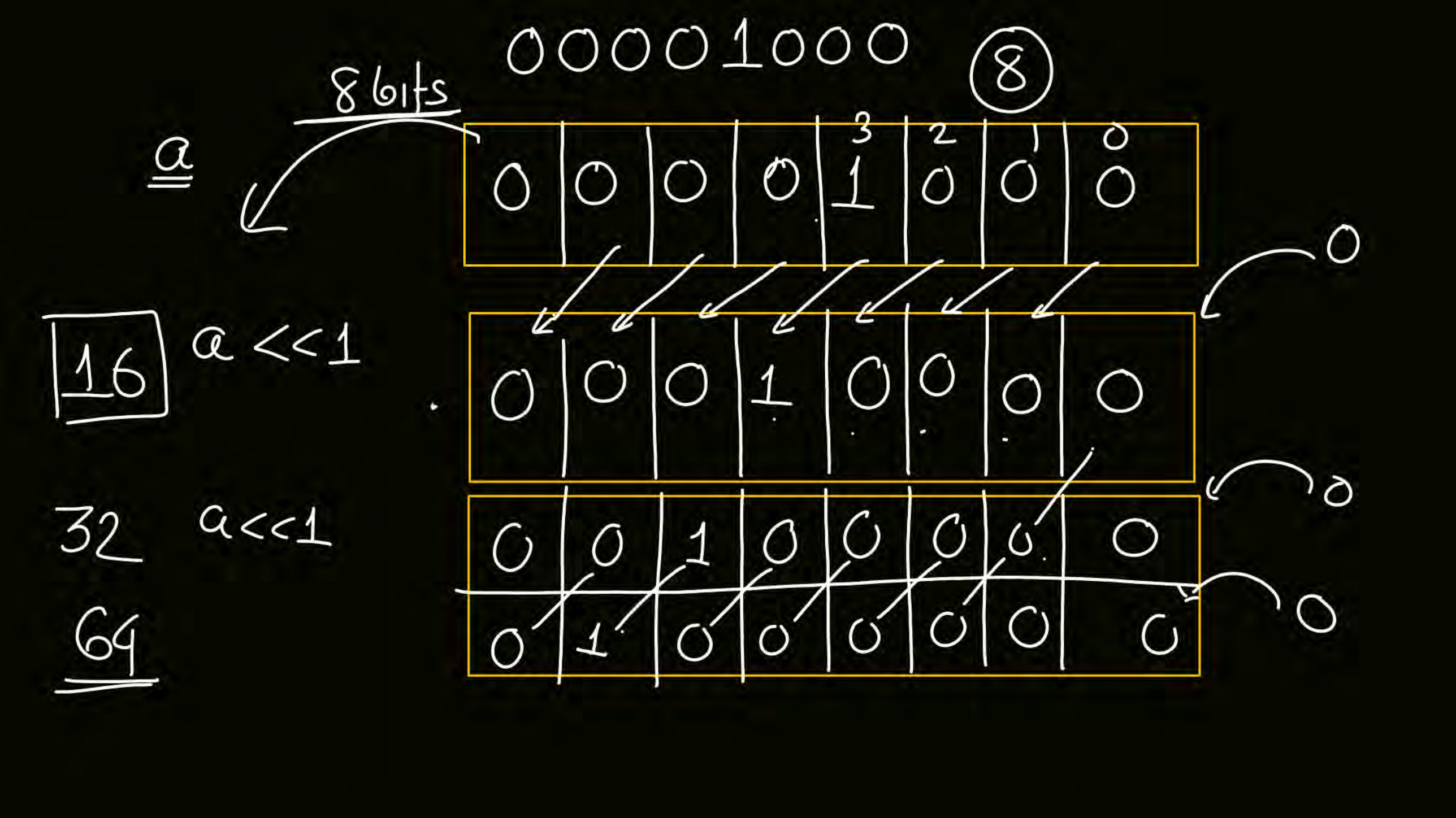
↑ times

Left shift a 3 times

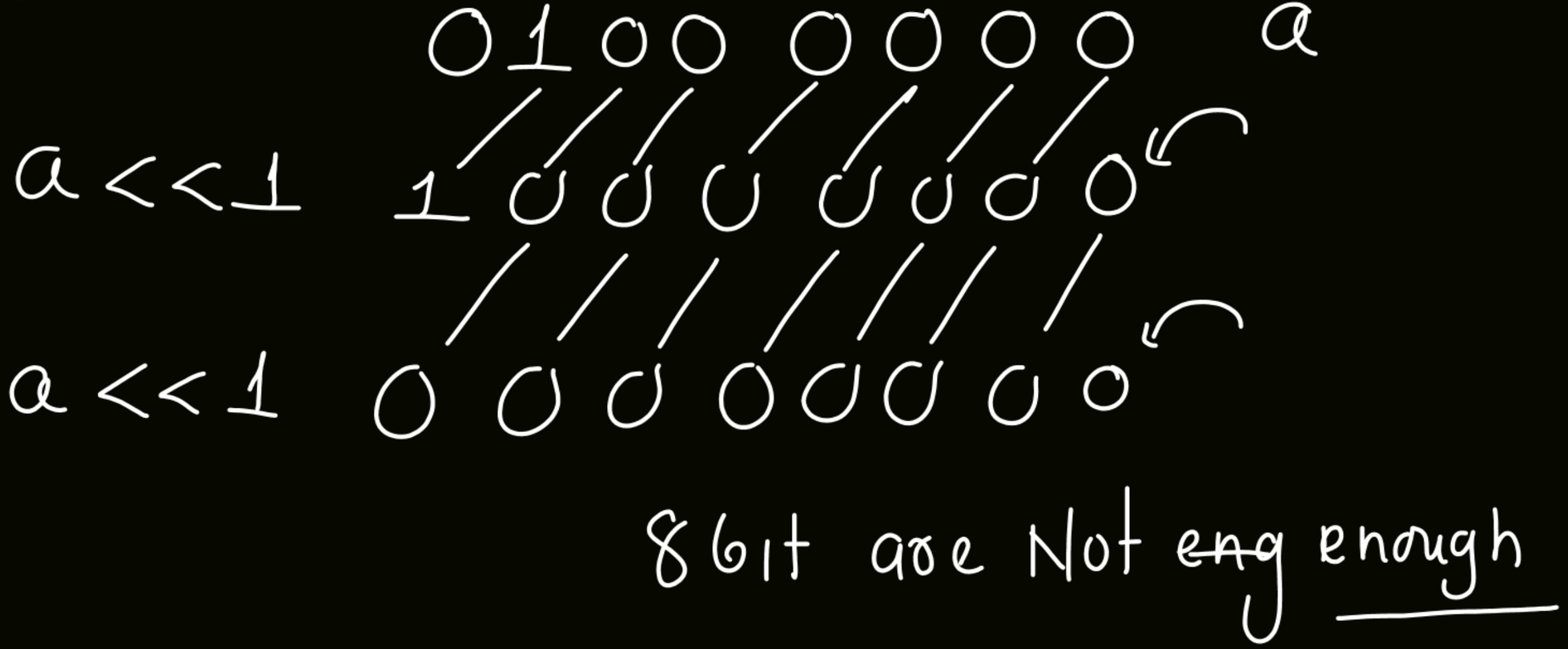
(A) 1 (B) 64

(C) 20 (D) -6

Slide


00001000
8
8 bits
a
3
2
1
0
0 0 0 0 1 0 0 0
16
a<<1
0
0 0 0 1 0 0 0 0
32
a<<1
0
0 0 1 0 0 0 0 0
64
0 1 0 0 0 0 0 0
0

G239484606



8 bit are Not ~~enq~~ enough

Each bit shift Left operation is equal to multiply by 2 if overflow does not occurs.



Bit-wise Operator

```
#include<stdio.h>
   int main(){
            char a = 8;
            int k;
            k =a<<3;
            printf("%d", k);
            return 0;
   }
```

(A) 1 **(B) 64**

(C) 20 (D) -6

Slide



Bit-wise Operator

```
#include<stdio.h>
   int main(){
            char a = 64;
            int k;
            k =a>>3;
            printf("%d", k);
            return 0;
   }
```

(A) 1 (B) 21

(C) 8 (D) -6

Slide



2 mins Summary

&pic

&pic

&pic

&pic

&pic

Slide

THANK - YOU

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